

A Certified Multi-Run Capacity Lower Bound for the Binary Deletion Channel at $d = 1/2$

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Abstract—The capacity of the binary deletion channel remains unknown even when the deletion probability is one half. For alternating renewal inputs, the exact information rate is the classical single-output-run term plus the entropy rate of the hidden segmentation between successive output runs. We prove that this segmentation rate is the nondecreasing limit of finite endpoint-window entropies and that every successive increment is a conditional mutual information between the two extreme boundaries of the window. For any deletion probability, each conditional segmentation law is proportional to products of integer binomial differences; at deletion probability one half, a dyadic run-length law makes the common channel factor dyadic and permits a fully rational certificate. Entropy-tree and log-sum inequalities, an exact leaf-absorption identity, a binary chain-rule decomposition, and directed fixed-point bounds reduce the required entropies to finite nonnegative sums. Evaluating these sums for an explicit dyadic run-length law with 84 positive masses proves $C(1/2) > 0.11032415$, a roughly six-percent improvement over the previously published lower bound 0.104075. Independent direct enumeration on small instances agrees with the directed evaluations and places every certified value below its numerical reference value. For context, a guarded rational reconstruction of the strongest published finite-channel table gives $C(1/2) < 0.189538767616$ under the stated table-precision assumption.

Index Terms—Deletion channels, capacity bounds, renewal processes, conditional entropy, certified computation.

I. INTRODUCTION

THE binary deletion channel with deletion probability d removes each transmitted bit independently with probability d and delivers the surviving bits in order, without marking where deletions occurred. Its capacity $C(d)$, in bits per transmitted bit, is well defined by Dobrushin’s coding theorem for synchronization channels [1], yet no single-letter expression is known for any $d \in (0, 1)$. The problem is a benchmark for channels with memoryless symbol loss but unbounded synchronization ambiguity; surveys of the principal techniques appear in [2], [3].

The half-deletion point $d = 1/2$ is canonical and quantitatively unresolved. On the lower side, Rubinstein and Con obtained

$$C(1/2) > 0.104075 \quad (1)$$

by optimizing run-length distributions inside the Drinea–Mitzenmacher framework [4]. On the upper side, Pinto and Ribeiro reported $C(1/2) < 0.1896$ from exact finite deletion channels evaluated with a parallelized Blahut–Arimoto algorithm [5]–[7]. The remaining gap is large for so elementary a channel.

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A. The information discarded by single-run decoding

The most successful lower-bound family uses an input whose binary symbols alternate in runs, with run lengths independent and identically distributed according to a law P . Drinea and Mitzenmacher grouped the source runs into *types*—the maximal segments that merge into one output run—and bounded the achievable rate by an expression that treats each output run in isolation [8]. Kirsch and Drinea later proved an exact information-rate formula for this input class [9]. Rewritten in the notation of this paper (Theorem 8), the exact rate is

$$\mathcal{R}(P) = R_{\text{run}}(P) + \lambda(P) e(P), \quad (2)$$

where R_{run} is the computable single-output-run bound, λ is the density of output runs per input bit, and $e(P) \geq 0$ is the entropy rate of the *hidden boundaries*: given the concatenated source runs and all output-run lengths, one still does not know where one type ends and the next begins. Single-run decoding forfeits exactly $\lambda e(P)$. The survey [2] identified a rigorous lower bound on this forfeited term as an open direction; numerical optimization within the single-run family had visibly saturated near 0.1042 at $d = 1/2$, so any further progress had to come from $e(P)$.

This paper supplies the missing bounding theory and applies it. The development is deliberately split: Sections II–VIII are purely deductive and hold for every finite-support run-length law; the computation in Section IX only *instantiates* finitely many terms of those bounds at one explicit witness distribution and verifies the resulting inequalities in exact rational arithmetic.

B. Contributions

1) *Endpoint-window hierarchy*: Let W_m be the word formed by the source runs of m consecutive types after erasing the type boundaries, let K_1^m be the m output-run lengths, and let $B_1, \dots, B_{m-1}$ be the hidden boundary positions. Write $g_m = H(B_{m-1} \mid W_m, K_1^m)$ for the entropy of the *last* boundary. Theorem 11 proves

$$g_{m+1} - g_m = I(B_1; B_m \mid W_{m+1}, K_1^{m+1}) \geq 0, \quad (3)$$

that $g_m \uparrow e(P)$, and hence that

$$e(P) \geq g_3 + \sum_{m=4}^M \Delta_m, \quad \Delta_m = g_m - g_{m-1}, \quad (4)$$

for every M . The point of (4) is architectural: the asymptotic quantity $e(P)$ is replaced by an *additive* family of finite-window terms, each a conditional entropy or conditional

mutual information of finitely many discrete variables, each nonnegative, and each certifiable in isolation. Partial progress on any term is never wasted and never double counted.

2) *Integer posterior and the half-deletion specialization:* For every deletion probability, all boundary placements compatible with a fixed source word and fixed output lengths have the same numbers of deleted and retained bits. Consequently (Lemma 13) their posterior probabilities are proportional to products of integer binomial differences and are *independent of both P and the deletion probability*. At deletion probability one half, the common factor is independent of the output lengths and is dyadic whenever P is dyadic. Every quantity in (4) then becomes a dyadically weighted sum of entropies of explicitly integer-weighted finite distributions for the witness used below.

3) *Entropy inequalities with exact remainders:* To bound the integer-weighted entropies from below without evaluating them exactly, we prove: an entropy-tree inequality (Lemma 17); log-sum superadditivity of the two-argument entropy functional F (Lemma 16); an exact leaf-absorption decomposition writing the entire tree-inequality gap as a sum of integrals $\int \log_2(1 + c/x) dx$ with convex integrands (Lemma 18); and an exact binary chain-rule expansion of the mutual information between multivalued endpoints into 2×2 tables (Lemma 20). Each device either is an identity or discards a provably nonnegative remainder, so truncation always acts in the safe direction.

4) *Directed fixed-point arithmetic:* Section VIII proves correctness of the arithmetic used to evaluate the surviving finite sums: dyadic floor rounding of source-word probabilities, two-sided integer $\log_2$ bounds derived from the inverse hyperbolic tangent series with certified rational constants, homogeneity-based scaling of F , and midpoint lower bounds for the convex leaf integrals. Every rounding is directed toward the final inequality.

5) *Certified numerical result:* For the dyadic witness distribution specified in Appendix D (denominator 2^{30} , 84 positive masses, and maximum run length 150), the directed finite evaluation proves the computer-assisted theorem

$$C(1/2) > 0.11032415. \quad (5)$$

The underlying rational evaluates to

$$0.11032415011972435980010110\dots, \quad (6)$$

so the decimal in (5) is rounded toward the interior of the certified region. Relative to (1) this is an improvement of about 6.0%. A guarded exact reconstruction of the public $n = 29$ finite-channel table (one full unit of the last printed decimal added to every entry) gives

$$C(1/2) < 0.189538767616, \quad (7)$$

whence the interval

$$0.11032415 < C(1/2) < 0.189538767616. \quad (8)$$

The upper inequality in (8) is conditional on the printed-table precision assumption stated in Section IX-G. It is not a new finite-channel computation, but a conservative rational restatement of the table in [5].

C. Relation to prior work

The single-run bound and its run-length optimization are due to [8] with numerical refinements in [4]. The exact rate formula for i.i.d.-run inputs is due to [9]; the survey [2] isolates the nonnegative correction and observes that joint decoding of several output runs would recover part of it, without giving a bounding method. Upper bounds via finite channels originate with [10] and were sharpened by [5]. The present paper contributes the monotone endpoint-window representation of the correction, the integer posterior and entropy-inequality toolkit that makes finite windows certifiable, and the first deterministic certificate that strictly exceeds the published single-run record at $d = 1/2$. The certificate is computer assisted in the sense of finite enumeration with directed rounding; it involves no sampling, floating-point transcendental evaluation, or stopping tolerance in the lower-bound comparison.

D. Organization

Section II defines the channel, the renewal input, and the integer type multiplicities. Section III states the exact rate decomposition and the computable baseline. Section IV proves the endpoint-window hierarchy. Section V establishes the integer posterior structure and the dyadic half-deletion specialization. Sections VI and VII derive finite lower bounds for g_3 and for the window increments Δ_m . Section VIII proves the directed-rounding rules. Section IX reports the numerical verification and the guarded upper reconstruction. Section X discusses limitations and continuations. The appendices give algorithmic details, bookkeeping proofs, the complete witness, and the exact ledger.

II. CHANNEL, RENEWAL INPUT, AND OUTPUT TYPES

A. Binary deletion channel

For a finite binary word $x = x_1 \cdots x_n$, the channel BDC_d retains each coordinate independently with probability $1 - d$ and outputs the subsequence of retained symbols in their original order. Let $Y(x)$ denote the random output. The information capacity is

$$C(d) = \lim_{n \rightarrow \infty} \frac{1}{n} \max_{P_{X^n}} I(X^n; Y(X^n)), \quad (9)$$

and by Dobrushin's theorem the limit exists and coincides with the operational capacity [1]. Moreover, the information rate of any stationary ergodic input process is achievable [1], [9], which is the only consequence of (9) used below. Throughout the remainder of the paper

$$d = \frac{1}{2}, \quad (10)$$

all logarithms are base two, and $0 \log 0 := 0$.

B. Alternating renewal input

Let $P = (P_x)_{x \geq 1}$ be a probability law with finite support. Generate run lengths

$$X_1, X_2, \dots \stackrel{\text{i.i.d.}}{\sim} P \quad (11)$$

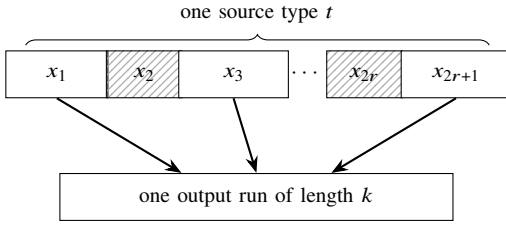


Fig. 1. One output run and its source type. The white odd-indexed runs carry the output symbol and feed the output run; the hatched even-indexed runs carry the opposite symbol and are deleted completely. The first odd-indexed run must contribute at least one surviving bit.

and, independently, a fair initial symbol. The input is the concatenation of binary runs of lengths $X_1, X_2, \dots$ with alternating symbols. Edge effects from the first and last partial runs are $O(1)$ in entropy and vanish after normalization by the input length; equivalently one may work with the stationary two-sided version of the process. Define

$$\mu = \mathbb{E}[X_1], \quad D = \mathbb{E}[2^{-X_1}] = \sum_{x \geq 1} P_x 2^{-x}. \quad (12)$$

Since a run of length x is fully deleted with probability 2^{-x} at $d = 1/2$, the quantity D is the probability that a given complete input run leaves no trace in the output.

An output run begins at an input run from which at least one bit survives, and it absorbs every later input run of the same symbol reachable by deleting all intervening opposite-symbol runs completely. The next subsection formalizes the resulting decomposition of the input into *types*; the density of output runs is then computed in Corollary 5.

C. Types and their integer multiplicities

Definition 1 (Type). A *type* is an odd-length source-run word

$$t = (x_1, x_2, \dots, x_{2r+1}), \quad r \geq 0, \quad (13)$$

recording the complete input runs consumed by one output run: the odd-indexed runs carry the output symbol, the r even-indexed runs carry the opposite symbol and are fully deleted, and at least one bit of the first run survives. Write

$$s(t) = \sum_{j=0}^r x_{2j+1}, \quad s_+(t) = s(t) - x_1 \quad (14)$$

for the total same-symbol length and its part beyond the first run.

The condition “at least one bit of the first run survives” is what makes the first run the *start* of the output run; the later odd runs may contribute any number of bits, including zero, because the output run has already started when they are reached. Figure 1 shows the correspondence.

Lemma 2 (Type multiplicity). Fix a type t and an integer $k \geq 1$. The number of deletion patterns of the same-symbol positions of t for which the output run has length exactly k and the first run contributes at least one bit is

$$A(t, k) = \binom{s(t)}{k} - \binom{s_+(t)}{k}. \quad (15)$$

Moreover $A(t, k) > 0$ if and only if $1 \leq k \leq s(t)$.

Proof. A deletion pattern producing output length k retains exactly k of the $s(t)$ same-symbol bits; there are $\binom{s(t)}{k}$ such subsets. Among them, exactly $\binom{s_+(t)}{k}$ retain no bit of the first run, since those choose all k survivors from the $s_+(t)$ later same-symbol positions. Subtracting yields (15).

For the support claim: if $k > s(t)$, both binomial coefficients vanish. If $1 \leq k \leq s(t)$ and $k > s_+(t)$, the second coefficient vanishes and the first is positive. If $1 \leq k \leq s_+(t)$, then $s_+(t) < s(t)$ (because $x_1 \geq 1$) and strict monotonicity of $n \mapsto \binom{n}{k}$ for $n \geq k$ gives $\binom{s(t)}{k} > \binom{s_+(t)}{k}$. $\square$

For a general deletion probability $d \in (0, 1)$, every pattern counted by $A(t, k)$ retains k bits and deletes all remaining bits of the type. Thus the joint law of one type and its output-run length is

$$q_d(t, k) = \left(\prod_{j=1}^{2r+1} P_{x_j} \right) d^{\sum_j x_j - k} (1-d)^k A(t, k). \quad (16)$$

At $d = 1/2$ this specializes to

$$q(t, k) = \left(\prod_{j=1}^{2r+1} P_{x_j} 2^{-x_j} \right) A(t, k). \quad (17)$$

The first factor in (17) is the probability of drawing the runs of t and of one specified deletion pattern; the integer $A(t, k)$ counts all admissible patterns. Equation (16) makes clear that equal weighting within fixed (t, k) does not require $d = 1/2$. The half-deletion point is special because, for dyadic P , the common factor becomes dyadic and independent of k .

Put $D_d = \sum_x P_x d^x$. Summing (16) over k and the coordinates of a type with parameter r gives $(1 - D_d) D_d^r$. The proof below repeats verbatim with (q_d, D_d) , so successive pairs are i.i.d. for every $d \in (0, 1)$; henceforth $d = 1/2$.

Lemma 3 (Normalization and renewal structure). The function q of (17) is a probability law on pairs (T, K) . The total mass of types with exactly r deleted opposite-symbol runs equals $(1 - D) D^r$. Distinct output runs have independent and identically distributed type/length pairs $(T_j, K_j)_{j \geq 1} \sim q$.

Proof. Fix t with parameter r and sum (15) over k :

$$\sum_{k \geq 1} A(t, k) = 2^{s(t)} - 2^{s_+(t)} = 2^{s_+(t)} (2^{x_1} - 1). \quad (18)$$

Multiplying by $\prod_j 2^{-x_j} = 2^{-s(t)} \prod_{j \text{ even}} 2^{-x_j}$ gives

$$(1 - 2^{-x_1}) \prod_{j \text{ even}} 2^{-x_j}. \quad (19)$$

Now attach the run-length probabilities and sum each coordinate independently: the first run contributes $\sum_x P_x (1 - 2^{-x}) = 1 - D$; each of the r even runs contributes $\sum_x P_x 2^{-x} = D$; each later odd run contributes $\sum_x P_x = 1$. The mass at parameter r is therefore $(1 - D) D^r$, and $\sum_{r \geq 0} (1 - D) D^r = 1$ because $D < 1$ (indeed $D \leq 1/2$).

For independence, consider the two-sided stationary version of the input and condition on the event S_1 that a fixed run, indexed 1, starts a type. A run starts a type if and only if it contains a survivor and the nearest earlier run containing a

survivor lies at odd distance (so that its symbol is opposite); the second condition concerns only earlier runs and their patterns, which are independent of everything from run 1 onward, while the first has probability $1 - D$. Now fix types $t_1, \dots, t_m$ with output lengths $k_1, \dots, k_m$ and let $N = |t_1| + \dots + |t_m|$. From run 1 onward, the event $\{(T_j, K_j) = (t_j, k_j), j \leq m\}$ states that runs $1, \dots, N$ have the lengths prescribed by the t_j ; that within each t_j the even-indexed runs are fully deleted and the same-symbol pattern retains k_j bits including at least one from the first run; and that run $N + 1$ contains a survivor, which is what terminates t_m . (No separate condition marks the starts of $t_2, \dots, t_m$: the survivor requirement on the first run of t_j is already counted inside $A(t_j, k_j)$, and a type always ends exactly at the next opposite-symbol run containing a survivor, all intermediate fully deleted runs being absorbed as zero-contribution coordinates of the current type.) Since run lengths and per-run deletion patterns are i.i.d. across runs, the probability of this event factorizes as

$$\left(\prod_{j=1}^m q(t_j, k_j)\right) (1 - D), \quad (20)$$

the trailing factor being the survivor probability of run $N + 1$. Dividing by the probability of the run-1 requirement in S_1 , namely $1 - D$, gives

$$\mathbb{P}((T_j, K_j) = (t_j, k_j), j \leq m \mid S_1) = \prod_{j=1}^m q(t_j, k_j), \quad (21)$$

which is the claimed i.i.d. structure. $\square$

Remark 4. Lemma 3 is the engine of everything that follows: the output process is a renewal sequence of i.i.d. “puzzle pieces” (T_j, K_j) , and every question about the input–output information rate becomes a question about this i.i.d. sequence. The single-run bound of [8] uses each piece alone; the correction studied in this paper quantifies how much the *joints between pieces* are worth.

Corollary 5 (Output-run density). *A type consumes on average $(1 + D)/(1 - D)$ complete input runs. Consequently, if M_n denotes the number of complete types among the first n input bits, then almost surely and in mean*

$$\frac{M_n}{n} \longrightarrow \lambda(P) = \frac{1 - D}{(1 + D)\mu}. \quad (22)$$

Proof. A type with parameter r consumes $2r + 1$ runs, and by Lemma 3 the parameter is geometric with $\mathbb{P}(r) = (1 - D)D^r$, so the mean consumption is $1 + 2D/(1 - D) = (1 + D)/(1 - D)$, finite because $D \leq 1/2$. The runs consumed by successive types are i.i.d. blocks; the strong law applied to run counts and to run lengths (mean μ per run) gives $n/M_n \rightarrow \mu(1 + D)/(1 - D)$ almost surely, and L^1 convergence follows from renewal-reward theory since all means are finite. Inverting yields (22). $\square$

III. EXACT RENEWAL RATE AND THE SINGLE-RUN BASELINE

A. Boundary variables

Let $(T_j, K_j)_{j \geq 1}$ be the i.i.d. type/length sequence of Lemma 3. For $m \geq 1$ concatenate the source-run words while erasing the type boundaries,

$$W_m = T_1 \parallel T_2 \parallel \dots \parallel T_m, \quad (23)$$

so that W_m is a single word of run lengths, and record the erased boundary positions

$$B_i = |T_1| + \dots + |T_i|, \quad 1 \leq i \leq m - 1. \quad (24)$$

Because the $|T_j|$ are odd and positive, the map

$$T_1^m \longleftrightarrow (W_m, B_1^{m-1}) \quad (25)$$

is a bijection: splitting W_m at the positions B_1^{m-1} recovers the types. Define the *boundary entropies*

$$h_m = H(B_1^{m-1} \mid W_m, K_1^m), \quad h_1 = 0. \quad (26)$$

Section IV proves that h_m/m converges; call the limit

$$e(P) = \lim_{m \rightarrow \infty} \frac{h_m}{m} \geq 0. \quad (27)$$

The quantity h_m is the residual uncertainty about *where the m jigsaw pieces were cut apart*, after the receiver is told the entire concatenated source-run word and all m output-run lengths. It is exactly the information forfeited when each output run is decoded in isolation.

B. Exact rate decomposition

We first record an identity that follows only from the i.i.d. type sequence and the bijection (25).

Lemma 6 (Type–boundary entropy identity). *The conditional entropy $H(T \mid K)$ is finite and, for every $m \geq 1$,*

$$mH(T \mid K) = H(W_m \mid K_1^m) + h_m. \quad (28)$$

In particular, $0 \leq h_m \leq mH(T \mid K)$.

Proof. The number $2r + 1$ of runs in T has the geometric law of Lemma 3. Conditional on r , the type is a word over the finite support of P , so $H(T) \leq H(r) + \mathbb{E}[2r + 1] \log_2 |\text{supp}(P)| < \infty$ and hence $H(T \mid K) < \infty$. Since the pairs (T_j, K_j) are i.i.d.,

$$H(T_1^m \mid K_1^m) = mH(T \mid K). \quad (29)$$

Applying the bijection $T_1^m \leftrightarrow (W_m, B_1^{m-1})$ and then the chain rule gives

$$\begin{aligned} mH(T \mid K) &= H(W_m, B_1^{m-1} \mid K_1^m) \\ &= H(W_m \mid K_1^m) + h_m, \end{aligned} \quad (30)$$

which proves the claim. $\square$

The exact information-rate formula of Kirsch and Drinea can now be imported without changing its boundary term. In their notation, a source group G is precisely one binary source segment represented here by T , and its output block L is represented by K . For a binary alternating input, the fair initial symbol is independent of all lengths and is also revealed by the corresponding binary block, so it contributes

no conditional-entropy term; their symbol-transition term is zero and $H(G | L) = H(T | K)$. The mean source-group length in input bits is $1/\lambda$ by Corollary 5. Given W_m , the bit length of the first type and the run-index boundary B_1 determine one another because all run lengths are positive. Their one-index shift in the finite window does not affect the limit. Thus their theorem specializes as follows.

Fact 7 (Kirsch–Drinea [9, Th. 1], specialized). *For every finite-support run-length law P , the alternating renewal input has information rate*

$$\mathcal{R}(P) = \frac{H(P)}{\mu} - \lambda H(T | K) + \lambda \eta(P), \quad (31)$$

where

$$\eta(P) = \lim_{m \rightarrow \infty} \eta_m, \quad \eta_m = H(B_1 | W_m, K_1^m), \quad (32)$$

and the sequence $(\eta_m)_{m \geq 2}$ is nondecreasing and bounded.

Theorem 8 (Boundary form of the exact rate). *For every finite-support run-length law P ,*

$$\mathcal{R}(P) = R_{\text{run}}(P) + \lambda e(P), \quad (33)$$

where

$$R_{\text{run}}(P) = \frac{H(P)}{\mu} - \lambda H(T | K) \quad (34)$$

and $H(T | K) < \infty$. In particular $C(1/2) \geq R_{\text{run}}(P) + \lambda e(P)$.

Proof. Expand h_m from the first boundary:

$$h_m = H(B_1 | W_m, K_1^m) + H(B_2^{m-1} | B_1, W_m, K_1^m). \quad (35)$$

The first term is η_m . Given B_1 , cutting W_m produces (T_1, W') with $W' = T_2 \parallel \dots \parallel T_m$, and subtracting B_1 relabels B_2^{m-1} as the complete boundary vector of the suffix window. The pair (T_1, K_1) is independent of that suffix, so the second term is h_{m-1} . Thus $h_m = h_{m-1} + \eta_m$. Theorem 11, proved from the same i.i.d. type law and Lemma 6, gives $h_m = h_{m-1} + g_m$. Consequently

$$\eta_m = H(B_1 | W_m, K_1^m) = H(B_{m-1} | W_m, K_1^m) = g_m. \quad (36)$$

Moreover, Theorem 11 gives $g_m \uparrow e(P)$. Hence $\eta(P) = e(P)$, and substitution in Fact 7 yields (33)–(34). Finally, Dobrushin’s theorem makes this stationary ergodic input rate achievable, so $C(1/2) \geq \mathcal{R}(P)$ [1]. $\square$

Setting $e(P)$ to zero in (33) recovers the single-output-run bound; since $h_m \geq 0$ termwise, $e(P) \geq 0$ and the single-run bound is never optimistic. The objective of Sections IV–VIII is a deterministic, finitely checkable lower bound on $e(P)$.

C. Computable single-run baseline

The certificate needs a certified lower bound on $R_{\text{run}}(P)$ itself. For this we import the evaluation formula of Drinea and Mitzenmacher, whose equality with (34) is established by Kirsch and Drinea.

Fact 9 (Drinea–Mitzenmacher [8, Th. 3]; identification with the single-run term in [9]). *Define*

$$L(z) = \sum_{x \geq 1} P_x \left(\frac{1+z}{2} \right)^x, \quad D = L(0), \quad (37)$$

let $\pi = (\pi_k)_{k \geq 1}$ be the output-run length law with probability generating function

$$\Pi(z) = \sum_{k \geq 1} \pi_k z^k = \frac{L(z) - D}{1 - D L(z)}, \quad (38)$$

let $Q_{r,i} = \mathbb{P}(X_1 + \dots + X_i = r)$ with $Q_{0,0} = 1$, and put $A_{z,r}(k) = \binom{z+r}{k} - \binom{r}{k}$. Then, with

$$S(P) = \sum_{i=0}^{\infty} D^i \sum_{z \geq 1} P_z \sum_{r \geq 0} Q_{r,i} 2^{-(z+r)} \times \sum_{k=1}^{z+r} A_{z,r}(k) \log_2 A_{z,r}(k), \quad (39)$$

the single-run term of Theorem 8 equals

$$R_{\text{run}}(P) = -1 + \lambda [H(\pi) + S(P)]. \quad (40)$$

Formula (40) is certifiable from below by finite computation, for two structural reasons. First, every summand of (39) is nonnegative—the multiplicities $A_{z,r}(k)$ are nonnegative integers and $u \log_2 u \geq 0$ for integers $u \geq 0$ —so truncating the series in i can only decrease it. Second, entropy decreases under merging of atoms: for any distribution and any partition of its atoms into groups, the entropy of the grouped distribution is no larger, by the grouping identity $H(\text{fine}) = H(\text{coarse}) + \sum_g p_g H(\text{within } g) \geq H(\text{coarse})$. Hence computing finitely many π_k exactly from (38) and merging the entire uncomputed tail into a single atom gives a lower bound on $H(\pi)$. Both reductions leave only finite rational sums, evaluated with the directed logarithm bounds of Section VIII.

For the witness distribution of Section IX, direct evaluation truncates (39) at $i = 6$, computes the first 200 coefficients of (38) exactly, merges the remaining mass ($< 1.4 \times 10^{-15}$) into one atom, and proves

$$R_{\text{run}}(P) > 0.1028000591453039341 \dots, \quad (41)$$

of which the final ledger retains the rational assertion

$$R_{\text{run}}(P) > 0.10280005. \quad (42)$$

IV. ENDPOINT-WINDOW REPRESENTATION OF THE BOUNDARY ENTROPY RATE

The boundary entropy rate $e(P)$ is defined through a limit and is not directly computable. This section replaces it, from below, by quantities involving a *single* boundary in a *finite* window. Two elementary facts about discrete entropy are used repeatedly and are recorded first.

Lemma 10 (Conditioning tools). *Let X, Y, Z be discrete random variables.*

- a) *If Z is independent of the pair (X, Y) , then $H(X | Y, Z) = H(X | Y)$.*
- b) *If φ is a bijection between the ranges of Y and Y' with $Y' = \varphi(Y)$, then $H(X | Y) = H(X | Y')$. If for each y the*

map $x \mapsto \psi_Y(x)$ is injective, then $H(\psi_Y(X) | Y) = H(X | Y)$.

Proof. a) Independence gives $\mathbb{P}(X = x | Y = y, Z = z) = \mathbb{P}(X = x | Y = y)$ whenever the conditioning event has positive probability, so the conditional entropies agree after averaging over (Y, Z) . b) Conditional entropy depends only on the conditional laws and the law of the conditioning variable, both invariant under relabeling by bijections; the second claim holds because an injection applied for each fixed y relabels the conditional law without changing its entropy. $\square$

For $m \geq 2$ define the m -run endpoint entropy

$$g_m = H(B_{m-1} | W_m, K_1^m), \quad (43)$$

the uncertainty of the last hidden boundary alone.

Theorem 11 (Endpoint-window hierarchy). *For every finite-support run-length law P :*

$$h_m = h_{m-1} + g_m, \quad m \geq 2, \quad (44)$$

$$g_{m+1} \geq g_m, \quad m \geq 2, \quad (45)$$

$$g_{m+1} - g_m = I(B_1; B_m | W_{m+1}, K_1^{m+1}), \quad m \geq 2. \quad (46)$$

The sequence g_m is bounded above by $2H(T | K)$; its limit exists and equals $e(P)$, and in particular the limit in (27) exists. Consequently

$$e(P) \geq g_m \quad \text{for every finite } m \geq 2. \quad (47)$$

Proof. Recursion (44). By the chain rule,

$$\begin{aligned} h_m &= H(B_{m-1} | W_m, K_1^m) \\ &\quad + H(B_1^{m-2} | B_{m-1}, W_m, K_1^m). \end{aligned} \quad (48)$$

The first term is g_m by definition. For the second, observe that

$$(B_{m-1}, W_m) \longleftrightarrow (W_{m-1}, T_m) \quad (49)$$

is a bijection: cutting W_m at position B_{m-1} produces the prefix W_{m-1} and the suffix T_m , and conversely $W_m = W_{m-1} \parallel T_m$ with $B_{m-1} = |W_{m-1}|$. By Lemma 10b the second term of (48) equals

$$H(B_1^{m-2} | W_{m-1}, K_1^{m-1}, T_m, K_m). \quad (50)$$

Now B_1^{m-2} and (W_{m-1}, K_1^{m-1}) are functions of (T_1^{m-1}, K_1^{m-1}) , which is independent of (T_m, K_m) by Lemma 3; Lemma 10a removes (T_m, K_m) from the conditioning, leaving exactly h_{m-1} .

Monotonicity (45). Conditioning reduces entropy:

$$\begin{aligned} g_{m+1} &= H(B_m | W_{m+1}, K_1^{m+1}) \\ &\geq H(B_m | B_1, W_{m+1}, K_1^{m+1}). \end{aligned} \quad (51)$$

Given B_1 , cut W_{m+1} at B_1 ; by the bijection $(B_1, W_{m+1}) \leftrightarrow (T_1, W')$ with $W' = T_2 \parallel \dots \parallel T_{m+1}$, and by the conditional relabeling $B_m \mapsto B_m - B_1$ (injective for each fixed B_1), Lemma 10b turns the right side of (51) into

$$H(B_m - B_1 | T_1, K_1, W', K_2^{m+1}). \quad (52)$$

The variable $B_m - B_1$ is the last boundary position inside the word W' , and $(B_m - B_1, W', K_2^{m+1})$ is a function of (T_2^{m+1}, K_2^{m+1}) , independent of (T_1, K_1) ; removing the latter

(Lemma 10a) and using that (T_2^{m+1}, K_2^{m+1}) has the same law as (T_1^m, K_1^m) yields exactly g_m . Thus

$$H(B_m | B_1, W_{m+1}, K_1^{m+1}) = g_m, \quad (53)$$

and (45) follows from (51).

Increment identity (46). By the definition of conditional mutual information and (53),

$$\begin{aligned} I(B_1; B_m | W_{m+1}, K_1^{m+1}) &= H(B_m | W_{m+1}, K_1^{m+1}) \\ &\quad - H(B_m | B_1, W_{m+1}, K_1^{m+1}) \\ &= g_{m+1} - g_m. \end{aligned} \quad (54)$$

Boundedness and convergence. Iterating (44) from $h_1 = 0$ gives

$$h_m = \sum_{j=2}^m g_j. \quad (55)$$

From (28), $h_m \leq mH(T | K)$. Fix $j \geq 2$; by monotonicity and (55),

$$jg_j \leq \sum_{\ell=j+1}^{2j} g_\ell \leq h_{2j} \leq 2jH(T | K), \quad (56)$$

so $g_j \leq 2H(T | K) < \infty$ by Lemma 6. A nondecreasing bounded sequence converges; write g_∞ for its limit. By (55), h_m/m is the Cesàro average of the g_j (with a vanishing correction from the missing $j = 1$ term) and converges to the same g_∞ . This proves that the limit defining $e(P)$ in (27) exists and equals $g_\infty = \sup_m g_m$, whence (47). $\square$

Remark 12 (Why the last boundary). The recursion (44) says that enlarging the window by one type adds exactly the entropy of the newly created boundary, conditioned on everything visible. Monotonicity says that this newest boundary is hardest to locate in the longest window: the more types stand to its left, the less the visible data pins it down. The limit of this single-boundary entropy is precisely the per-run rate $e(P)$ —an instance of the general principle that the entropy rate of a stationary sequence equals the limit of one-symbol conditional entropies, here materialized with completely explicit finite conditioning.

Theorem 11 produces the additive hierarchy announced in (4):

$$\begin{aligned} e(P) &\geq g_M = g_3 + \sum_{m=4}^M \Delta_m, \\ \Delta_m &= I(B_1; B_{m-1} | W_m, K_1^m). \end{aligned} \quad (57)$$

Every Δ_m is a distinct nonnegative quantity, so lower bounds on g_3 and on each Δ_m —however partial—always add up to a valid lower bound on $e(P)$. Figure 2 summarizes the structure. The next three sections make each term finitely certifiable.

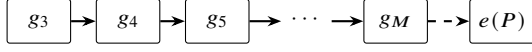


Fig. 2. The endpoint-window hierarchy. The solid arrow from g_{m-1} to g_m adds $\Delta_m \geq 0$; the dashed arrow denotes $g_M \uparrow e(P)$.

V. INTEGER POSTERIOR STRUCTURE AND THE HALF-DELETION SPECIALIZATION

The quantities g_3 and Δ_m are conditional entropies of the boundary placement given the observation (W_m, K_1^m) . This section shows that for every $d \in (0, 1)$ each such posterior is a normalized family of *integers*, independent of the run-length law P and of d . At $d = 1/2$ a dyadic P makes the common observation factor dyadic, so the complete finite evaluation can remain rational.

A. Segmentation states

Fix a window length $m \geq 2$ and condition on $W_m = w$ and $K_1^m = k_1^m$. Writing $r_j = (|T_j| - 1)/2$ for the parameter of the j th type, the word length constraint $|w| = m + 2R$ determines the total

$$R = r_1 + \dots + r_m, \quad (58)$$

so the possible boundary placements are indexed by the compositions

$$\mathcal{S}_{m,R} = \left\{ \mathbf{r} = (r_1, \dots, r_m) \in \mathbb{N}_0^m : \sum_j r_j = R \right\}. \quad (59)$$

Each state $\mathbf{r}$ splits w into odd-length type words $t_1(\mathbf{r}), \dots, t_m(\mathbf{r})$, namely $t_1(\mathbf{r})$ consists of the first $2r_1 + 1$ coordinates, $t_2(\mathbf{r})$ of the next $2r_2 + 1$, and so on; conversely the state determines the boundaries B_1^{m-1} . Define the integer weight

$$a_{\mathbf{r}}(w, k_1^m) = \prod_{j=1}^m A(t_j(\mathbf{r}), k_j) \in \mathbb{N}_0 \quad (60)$$

and, for a general deletion probability, the common observation factor

$$p_d(w, k_1^m) = \left(\prod_{x \in w} P_x \right) d^{\sum_{x \in w} x - \sum_j k_j} (1-d)^{\sum_j k_j}. \quad (61)$$

At $d = 1/2$ this factor is independent of k_1^m and becomes the word factor

$$p(w) = \prod_{x \in w} P_x 2^{-x}. \quad (62)$$

which is dyadic when P is dyadic.

Lemma 13 (Integer posterior). *For every $d \in (0, 1)$, window length m , word w with $|w| = m + 2R$, and output lengths k_1^m ,*

$$\mathbb{P}_d(T_1^m = (t_j(\mathbf{r}))_j, K_1^m = k_1^m) = p_d(w, k_1^m) a_{\mathbf{r}}(w, k_1^m). \quad (63)$$

Consequently, whenever $p_d(w, k_1^m) > 0$ and $Z(w, k_1^m) = \sum_{\mathbf{r} \in \mathcal{S}_{m,R}} a_{\mathbf{r}} > 0$,

$$\mathbb{P}(\text{state } \mathbf{r} \mid W_m = w, K_1^m = k_1^m) = \frac{a_{\mathbf{r}}(w, k_1^m)}{Z(w, k_1^m)}, \quad (64)$$

which does not depend on P or d . At $d = 1/2$, the factor in (63) is $p(w)$.

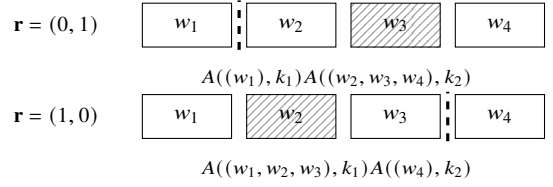


Fig. 3. Two segmentations of the same two-output-run observation for $R = 1$. Hatched runs are forced to be fully deleted, and the dashed line is the hidden type boundary. The posterior odds are the ratio of the two integer multiplicity products shown below the rows.

Proof. The left side of (63) is the product of the m factors (16). Across every segmentation, the run-length probabilities range over all coordinates of w , the total number of retained bits is $\sum_j k_j$, and the total number of deleted bits is $\sum_{x \in w} x - \sum_j k_j$. Their product is therefore the state-independent factor $p_d(w, k_1^m)$. The remaining factors are the multiplicities, whose product is $a_{\mathbf{r}}$. Equation (64) is Bayes' rule: for fixed k_1^m , the events $\{T_1^m = (t_j(\mathbf{r}))_j\}$ with $\mathbf{r} \in \mathcal{S}_{m,R}$ are exactly the ways of realizing $\{W_m = w\}$, and the common factor cancels from the ratio. $\square$

Remark 14 (What is special about $d = 1/2$). Integer posterior odds are not unique to half deletion: after conditioning on (w, k_1^m) , every placement has the same total numbers of deleted and retained bits. What is special at $d = 1/2$ is the common factor $p(w) = \prod_x P_x 2^{-x}$: for a dyadic P it is itself dyadic and contains no output-length-dependent powers of a non-dyadic channel parameter. The analytical posterior identities remain valid for general d ; the purely dyadic certificate below specializes to $d = 1/2$.

Figure 3 shows the smallest nontrivial example: the same observed data supports two boundary placements whose posterior odds are a ratio of integers.

B. Unnormalized entropy notation

For a finite family of nonnegative weights $z = (z_v)$ with total $Z_z = \sum_v z_v$, define

$$\mathcal{H}(z) = Z_z \log_2 Z_z - \sum_v z_v \log_2 z_v = Z_z H(z/Z_z), \quad (65)$$

with $\mathcal{H}(z) = 0$ if $Z_z = 0$. Working with $\mathcal{H}$ rather than H postpones all divisions: the certified objects remain integer weighted until the final rational comparison.

With this notation the endpoint entropy has a fully explicit form. For the three-run window, the last boundary B_2 is a function of the state coordinate r_3 , and conversely r_3 determines B_2 given w ; so, defining the *coarse endpoint weights*

$$u_c = \sum_{\substack{\mathbf{r} \in \mathcal{S}_{3,R} \\ r_3=c}} a_{\mathbf{r}}, \quad 0 \leq c \leq R, \quad (66)$$

the decomposition of g_3 over observations reads

$$g_3 = \sum_{(w, k_1^3)} p(w) \mathcal{H}(u_0, \dots, u_R). \quad (67)$$

Indeed $\mathbb{P}(W_3 = w, K_1^3 = k_1^3) = p(w)Z$ by Lemma 13, the posterior law of r_3 is $(u_c/Z)_c$, and $ZH(u/Z) = \mathcal{H}(u)$. The

sum converges: on the stratum with total parameter R the endpoint posterior has entropy at most $\log_2(R+1)$, so the stratum contributes at most $\mathbb{P}(r_1+r_2+r_3=R) \log_2(R+1)$, which is summable because the three independent type parameters have geometric laws. Every certified bound below proceeds by restricting the outer sum in (67)—and its analogues for Δ_m —to finite families of observations and bounding $\mathcal{H}$ from below pointwise; all omitted observations contribute nonnegative amounts.

VI. THREE-RUN ENDPOINT ENTROPY

This section derives finitely computable lower bounds for the pointwise entropies $\mathcal{H}(u_0, \dots, u_R)$ in (67) and assembles them into a bound on g_3 . All statements hold for arbitrary nonnegative weights and arbitrary finite-support P ; no numerical choice is made before Section IX.

A. The two-argument entropy functional

For $a, b \geq 0$ define

$$F(a, b) = (a+b) \log_2(a+b) - a \log_2 a - b \log_2 b, \quad (68)$$

the unnormalized entropy of a two-atom family: $F(a, b) = (a+b)H_2(\frac{a}{a+b})$ where H_2 is the binary entropy function.

Lemma 15 (Properties of F). *On $[0, \infty)^2$ the functional F is nonnegative, symmetric, continuous, homogeneous of degree one ($F(\gamma a, \gamma b) = \gamma F(a, b)$ for $\gamma \geq 0$), vanishes iff $a = 0$ or $b = 0$, and is nondecreasing in each argument, with*

$$\frac{\partial F}{\partial a}(a, b) = \log_2\left(1 + \frac{b}{a}\right) > 0 \quad (a, b > 0). \quad (69)$$

Proof. Nonnegativity, symmetry, continuity (with the convention $0 \log 0 = 0$), homogeneity, and the vanishing condition are immediate from $F = (a+b)H_2(a/(a+b))$ and $H_2 \in [0, 1]$, $H_2(0) = H_2(1) = 0$. Differentiating (68) in a gives $\log_2(a+b) + \frac{1}{\ln 2} - \log_2 a - \frac{1}{\ln 2} = \log_2 \frac{a+b}{a}$, which is positive; monotonicity extends to the boundary by continuity, using $F(0, b) = 0 \leq F(a, b)$. $\square$

Lemma 16 (Log-sum superadditivity). *For finitely many pairs $(a_i, b_i) \in [0, \infty)^2$,*

$$F\left(\sum_i a_i, \sum_i b_i\right) \geq \sum_i F(a_i, b_i). \quad (70)$$

Proof. Write $c_i = a_i + b_i$, $A = \sum a_i$, $B = \sum b_i$, $C = A + B$. Rearranging (68), inequality (70) is equivalent to

$$\sum_i a_i \log_2 \frac{a_i}{c_i} + \sum_i b_i \log_2 \frac{b_i}{c_i} \geq A \log_2 \frac{A}{C} + B \log_2 \frac{B}{C}, \quad (71)$$

and each of the two sums dominates the corresponding right-hand term by the log-sum inequality [11, Th. 2.7.1] applied to (a_i) against (c_i) and to (b_i) against (c_i) . Terms with $a_i = 0$ or $b_i = 0$ are covered by the convention $0 \log 0 = 0$. $\square$

B. Entropy-tree inequality and its exact gap

Lemma 17 (Entropy-tree inequality). *Let $z = (z_v)_{v \in V}$ be finite nonnegative weights and let $\mathcal{T}$ be a tree with vertex set V . Then*

$$\mathcal{H}(z) \geq \sum_{\{u,v\} \in E(\mathcal{T})} F(z_u, z_v). \quad (72)$$

Proof. Induct on $|V|$; the case $|V| = 1$ reads $\mathcal{H}(z) \geq 0$. Let ℓ be a leaf of $\mathcal{T}$ with neighbor v . Merging ℓ into v produces weights z' with $z'_v = z_v + z_\ell$, all other entries unchanged, on the contracted tree $\mathcal{T}' = \mathcal{T}/\ell$. Cancellation of the total-weight terms in (65) gives the exact identity

$$\mathcal{H}(z) = F(z_\ell, z_v) + \mathcal{H}(z'), \quad (73)$$

because $Z_z = Z_{z'}$ while the atom terms differ by $z_\ell \log z_\ell + z_v \log z_v - (z_\ell + z_v) \log(z_\ell + z_v) = -F(z_\ell, z_v)$. By induction, $\mathcal{H}(z') \geq \sum_{e \in E(\mathcal{T}')} F_e(z')$. Every edge of $\mathcal{T}'$ not incident to v carries the same weights as in $\mathcal{T}$; on an edge $\{v, u\}$, Lemma 15 gives $F(z_v + z_\ell, z_u) \geq F(z_v, z_u)$. Adding the removed leaf edge $F(z_\ell, z_v)$ proves (72). $\square$

The inequality (72) is generally strict, and the next lemma expresses its *entire* gap as a sum of explicit nonnegative integrals. This matters because the gap is not a nuisance term: for the segmentation posteriors it carries a substantial fraction of the entropy, and an exact positive decomposition allows any portion of it to be reclaimed safely.

Lemma 18 (Leaf-absorption decomposition). *For weights z and a tree $\mathcal{T}$ define the gap $\Delta_{\mathcal{T}}(z) = \mathcal{H}(z) - \sum_{\{u,v\} \in E(\mathcal{T})} F(z_u, z_v) \geq 0$. Let ℓ be a leaf with neighbor v and let $z', \mathcal{T}'$ be as in the proof of Lemma 17. Then*

$$\Delta_{\mathcal{T}}(z) = \Delta_{\mathcal{T}'}(z') + \sum_{u \in N_{\mathcal{T}}(v) \setminus \{\ell\}} [F(z_v + z_\ell, z_u) - F(z_v, z_u)], \quad (74)$$

and each bracket admits the integral representation

$$F(b+a, c) - F(b, c) = \int_b^{b+a} \log_2\left(1 + \frac{c}{x}\right) dx \geq 0 \quad (75)$$

with $a = z_\ell$, $b = z_v$, $c = z_u$. Iterating (74) until one vertex remains writes $\Delta_{\mathcal{T}}(z)$ exactly as a finite sum of such integrals. The representation (75) remains valid for $b = 0$, the singularity of the integrand being integrable.

Proof. Subtract the edge sums of $\mathcal{T}$ and $\mathcal{T}'$: the leaf edge contributes $F(z_\ell, z_v)$, which cancels against (73); edges not incident to v are unchanged; each remaining edge $\{v, u\}$ changes from $F(z_v, z_u)$ to $F(z_v + z_\ell, z_u)$. This proves (74). The fundamental theorem of calculus with (69) proves (75) for $b > 0$; for $b = 0$ both sides are $F(a, c)$, the integral converging since $\int_0^a \log_2(1 + c/x) dx \leq \int_0^a (\log_2(x+c) - \log_2 x) dx < \infty$. The terminal tree has one vertex and zero gap, so the iteration is exact. $\square$

C. The fine adjacent path and its expectation

Return to the three-run window with total parameter $R \geq 1$ and abbreviate, for a fixed observation (w, k_1^3) ,

$$z_{a,c} = a_{(a, R-a-c, c)}(w, k_1^3), \quad a, c \geq 0, \quad a+c \leq R, \quad (76)$$

so that $u_c = \sum_{a=0}^{R-c} z_{a,c}$. Applying Lemma 17 to the path $0-1-\dots-R$ on the endpoint values and then Lemma 16 to each edge, splitting by the common value of $a = r_1$,

$$\begin{aligned} \mathcal{H}(u_0, \dots, u_R) &\geq \sum_{c=1}^R F(u_{c-1}, u_c) \\ &\geq \sum_{c=1}^R \sum_{a=0}^{R-c} F(z_{a,c-1}, z_{a,c}). \end{aligned} \quad (77)$$

Zero weights require no special treatment: $F(0, \cdot) = 0$.

Each fine edge in (77) compares two states that differ by moving the last boundary across one deleted pair, with the first type untouched. Its expectation therefore factorizes into a mass factor for the first type and a purely local quantity for the last two types. To formalize this, fix integers $s \geq 1$ and $0 \leq i < s$, let $x = (x_1, \dots, x_{2s+2})$ be a local source word, and define the two adjacent cut pairs

$$(t_L^{(i)}, t_R^{(i)}) = (x_1^{2i+1}, x_{2i+2}^{2s+2}), \quad (78)$$

$$(t_L^{(i+1)}, t_R^{(i+1)}) = (x_1^{2i+3}, x_{2i+4}^{2s+2}), \quad (79)$$

where $x_a^b = (x_a, \dots, x_b)$, together with the nonnegative local edge sum

$$\begin{aligned} \mathcal{J}_{s,i}(P) &= \sum_{x_1, \dots, x_{2s+2} \geq 1} \left(\prod_{j=1}^{2s+2} P_{x_j} 2^{-x_j} \right) \\ &\times \sum_{k, \ell \geq 1} F(A(t_L^{(i)}, k) A(t_R^{(i)}, \ell), \\ &\quad A(t_L^{(i+1)}, k) A(t_R^{(i+1)}, \ell)), \end{aligned} \quad (80)$$

finite for finite-support P .

Proposition 19 (Three-run path bound). *For every integer $R_{\max} \geq 1$,*

$$g_3 \geq \sum_{R=1}^{R_{\max}} (1-D) \sum_{s=1}^R D^{R-s} \sum_{i=0}^{s-1} \mathcal{J}_{s,i}(P). \quad (81)$$

The inequality remains valid if any $\mathcal{J}_{s,i}$ is replaced by the sub-sum of (80) over any finite set of local words.

Proof. Insert (77) into (67) and keep only $1 \leq R \leq R_{\max}$; the discarded strata are nonnegative. Fix R , an edge index pair (a, c) with $1 \leq c \leq R-a$, and consider its total contribution

$$\sum_{(w, k_1^3): |w|=3+2R} p(w) F(z_{a,c-1}, z_{a,c}). \quad (82)$$

Both states appearing in the edge have first coordinate a , hence share the same first type word t_1 , namely the first $2a+1$ coordinates of w , and the same factor $A(t_1, k_1)$ in their weights (60). By homogeneity (Lemma 15),

$$F(z_{a,c-1}, z_{a,c}) = A(t_1, k_1) F(\alpha \beta_{c-1}, \alpha' \beta_c), \quad (83)$$

where, writing $x = (x_1, \dots, x_{2s+2})$ for the last $2s+2$ coordinates of w with $s = R-a$ and $i = s-c$: $\alpha = A(t_L^{(i+1)}, k_2)$, $\beta_{c-1} = A(t_R^{(i+1)}, k_3)$ are the multiplicities of the state with $r_3 = c-1$ (whose middle type has parameter $i+1$), and $\alpha' = A(t_L^{(i)}, k_2)$, $\beta_c = A(t_R^{(i)}, k_3)$ those of the state with $r_3 = c$. Also $p(w)$ factors as the dyadic probability of the first $2a+1$ coordinates

times that of x . Summing (82) first over the first type's word and k_1 gives, by Lemma 3, the total mass $(1-D)D^a = (1-D)D^{R-s}$ of types with parameter a ; summing then over x and (k_2, k_3) gives exactly $\mathcal{J}_{s,i}(P)$, with F 's symmetry absorbing the order of arguments. As (a, c) ranges over $\{(a, c) : a \geq 0, 1 \leq c \leq R-a\}$, the pair $(s, i) = (R-a, s-c)$ ranges over $\{1 \leq s \leq R, 0 \leq i \leq s-1\}$ bijectively. Summing over $R \leq R_{\max}$ yields (81). Restricting any $\mathcal{J}_{s,i}$ to finitely many local words discards nonnegative summands only. $\square$

D. Endpoint corrections beyond the fine path

The fine path (77) is not all of $\mathcal{H}(u)$. Define, pointwise in the observation,

$$\Gamma_R = \Gamma_R^{\text{tree}} + \Gamma_R^{\text{logsum}}, \quad (84)$$

$$\Gamma_R^{\text{tree}} = \mathcal{H}(u) - \sum_{c=1}^R F(u_{c-1}, u_c), \quad (85)$$

$$\Gamma_R^{\text{logsum}} = \sum_{c=1}^R \left[F(u_{c-1}, u_c) - \sum_{a=0}^{R-c} F(z_{a,c-1}, z_{a,c}) \right]. \quad (86)$$

By Lemmas 17 and 16, $\Gamma_R^{\text{tree}} \geq 0$ and every bracket of (86) is nonnegative; by construction,

$$\mathcal{H}(u) = \sum_{c=1}^R \sum_{a=0}^{R-c} F(z_{a,c-1}, z_{a,c}) + \Gamma_R \quad (87)$$

exactly. Hence any lower bound on the expectation $\mathbb{E}[\Gamma_R]$ —by which we always mean the explicitly weighted sum $\sum_{(w, k_1^3)} p(w) \Gamma_R(w, k_1^3)$ restricted to a finite observation family—adds to Proposition 19 without double counting: the two parts partition $\mathcal{H}(u)$ observation by observation.

The two parts are certified differently, and the tools are already in place:

- i) Γ_R^{tree} is evaluated by Lemma 18: absorbing the path's leaves one at a time writes the gap as a finite sum of integrals (75), and each integral is bounded below by the midpoint rule of Lemma 26 because the integrand is convex. Several deterministic absorption orders may be tried, and the pointwise maximum of the resulting lower bounds retained: a maximum of valid lower bounds is a valid lower bound.
- ii) Γ_R^{logsum} is certified bracket by bracket as the difference of the directed evaluations $F(u_{c-1}, u_c)$ and $\sum_a \bar{F}(z_{a,c-1}, z_{a,c})$, where $\underline{F} \leq F \leq \bar{F}$ are supplied by Section VIII; if directed rounding drives a bracket negative it is replaced by zero, which is valid since the true bracket is nonnegative.

Appendix B records the bookkeeping in full.

For $R = 1$ the correction collapses to a closed form worth stating: the states are $(0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$ with weights $(z_0, z_1, z_2) = (z_{0,1}, z_{0,0}, z_{1,0})$, the endpoint weights are $(u_0, u_1) = (z_1 + z_2, z_0)$, and

$$\begin{aligned} \Gamma_1 &= F(z_0, z_1 + z_2) - F(z_0, z_1) \\ &= \mathcal{H}(z_0, z_1, z_2) - F(z_0, z_1) - F(z_1, z_2). \end{aligned} \quad (88)$$

The first equality is (86) with the single bracket $c = 1$ (here $\Gamma_1^{\text{tree}} = 0$ since u has two atoms); the second follows from two

applications of (73), absorbing z_2 into z_1 and then observing $\mathcal{H}(z_0, z_1+z_2) = F(z_0, z_1+z_2)$. Thus for $R = 1$ the general three-state tree-gap evaluation computes the endpoint correction *exactly* in the form (88), before directed rounding is applied.

VII. HIGHER ENDPOINT WINDOWS

By Theorem 11 it remains to bound the increments $\Delta_m = I(B_1; B_{m-1} \mid W_m, K_1^m)$ from below. This section reduces each Δ_m to finite integer posterior tables and gives an exact expansion of the multivalued endpoint mutual information into 2×2 blocks, each certifiable by directed rounding.

A. Endpoint variables and pointwise reduction

Work in the m -run window of Section V. Given $W_m = w$, the first boundary determines and is determined by the first coordinate of the state, $B_1 = 2r_1 + 1$, and likewise $B_{m-1} = |w| - (2r_m + 1)$; mutual information is invariant under such conditional relabelings (Lemma 10b applied inside each conditioning event). Hence, writing $A = r_1$ and $D_m = r_m$,

$$\Delta_m = \sum_{(w, k_1^m)} p(w) \mathcal{I}[\text{joint table of } (A, D_m)], \quad (89)$$

where, for a nonnegative matrix $q = (q_{\alpha\delta})$ with total Z_q , the *unnormalized mutual information* is

$$\mathcal{I}(q) = Z_q \mathcal{I}(\alpha; \delta) = \sum_{\alpha, \delta} q_{\alpha\delta} \log_2 \frac{q_{\alpha\delta} Z_q}{q_{\alpha\cdot} q_{\cdot\delta}}, \quad (90)$$

and the table entry at (α, δ) is $\sum_{\mathbf{r}: r_1=\alpha, r_m=\delta} a_{\mathbf{r}}(w, k_1^m)$, an integer by Lemma 13. Restricting the outer sum of (89) to any family of source words keeps a valid lower bound, because $\mathcal{I} \geq 0$ pointwise.

B. Exact binary chain-rule decomposition

The endpoint variables take values in $\{0, \dots, R\}$, so a single binary coarsening ϕ, ψ of the two endpoints, combined with the data-processing inequality $I(A; D \mid O) \geq I(\phi(A); \psi(D) \mid O)$, loses information. The certificate instead recovers the full multivalued mutual information through binary digit expansions.

Lemma 20 (Binary chain rule). *Let A, D take values in $\{0, \dots, R\}$, let $L = \lceil \log_2(R+1) \rceil$, and encode $A \mapsto A_1^L$, $D \mapsto D_1^L$ by any fixed injective bit maps. Then, for any conditioning variable O ,*

$$I(A; D \mid O) = \sum_{i=1}^L \sum_{j=1}^L I(A_i; D_j \mid O, A_1^{i-1}, D_1^{j-1}). \quad (91)$$

Proof. Injectivity gives $I(A; D \mid O) = I(A_1^L; D_1^L \mid O)$, since mutual information is invariant under injective relabelings of each argument. Expand first over the A bits by the chain rule for mutual information,

$$I(A_1^L; D_1^L \mid O) = \sum_{i=1}^L I(A_i; D_1^L \mid O, A_1^{i-1}), \quad (92)$$

then expand each term over the D bits with conditioning (O, A_1^{i-1}) held fixed. Both expansions are instances of the standard chain rule [11, Th. 2.5.2]. $\square$

Every summand of (91) is the mutual information of a 2×2 table, obtained by restricting the integer endpoint table to one prefix event $(A_1^{i-1}, D_1^{j-1}) = (\text{fixed})$ and aggregating by the bits (A_i, D_j) . The next lemma expresses it entirely through F , which is the form the directed arithmetic evaluates.

Lemma 21 (2×2 form). *For a nonnegative 2×2 table $q = \begin{pmatrix} q_{00} & q_{01} \\ q_{10} & q_{11} \end{pmatrix}$ with column sums $c_j = q_{0j} + q_{1j}$,*

$$\mathcal{I}(q) = F(c_0, c_1) - F(q_{00}, q_{01}) - F(q_{10}, q_{11}). \quad (93)$$

Proof. With Z_q the total, $Z_q \mathcal{I}(\alpha; \delta) = Z_q [H(\delta) - H(\delta \mid \alpha)]$. The unnormalized entropy of the column variable is $\mathcal{H}(c_0, c_1) = F(c_0, c_1)$, and the unnormalized conditional entropy given the row variable is the sum of the rows' unnormalized entropies, $\mathcal{H}(q_{00}, q_{01}) + \mathcal{H}(q_{10}, q_{11}) = F(q_{00}, q_{01}) + F(q_{10}, q_{11})$; two-atom families satisfy $\mathcal{H} = F$ by definition. $\square$

The certified evaluation of (93) uses a lower bound for the positive term and upper bounds for the two subtracted terms (Section VIII); if the directed result is negative it is clipped to zero, which is valid because $\mathcal{I} \geq 0$. Summing over all prefixes and digit pairs then reconstitutes, from below, the exact multivalued endpoint mutual information of each observation—not merely one binary projection of it.

C. Exhaustive output-length enumeration

For each source word a finite enumeration must visit every output-length vector that can contribute. The following support lemma gives an a priori computable exact range.

Lemma 22 (Second-largest cutoff). *Fix a source word and a set $\mathcal{A}$ of segmentation states still admissible after output lengths k_1^{j-1} have been fixed (that is, of positive weight so far). For $\mathbf{r} \in \mathcal{A}$ put $s_j(\mathbf{r}) = s(t_j(\mathbf{r}))$, the same-symbol total of the j th type word under $\mathbf{r}$. Define the second-largest value to be zero when fewer than two states remain. If k_j exceeds the second-largest element of the multiset $\{s_j(\mathbf{r}) : \mathbf{r} \in \mathcal{A}\}$, then after fixing k_j at most one state retains positive weight, and the observation contributes zero to the endpoint mutual information and to every entropy gap, whatever k_{j+1}^m may be.*

Proof. By Lemma 2, $A(t, k) > 0$ iff $1 \leq k \leq s(t)$; thus state $\mathbf{r}$ survives coordinate j iff $k_j \leq s_j(\mathbf{r})$. If k_j exceeds the second-largest total, only a state attaining the unique largest total can survive. A posterior supported on at most one state makes the boundary placement deterministic: all conditional entropies of functions of the state vanish, and so do all mutual informations. Later coordinates only multiply weights by further $A(\cdot, k)$ factors, which can remove states but never revive one, so the contribution remains zero for every continuation. $\square$

Consequently a recursive enumeration that ranges each k_j from 1 to the second-largest admissible total among the currently surviving states visits *every* observation with a possibly positive contribution; the suppressed continuations contribute exactly zero. This is an exactness statement, not an approximation: within a fixed source-word family, the enumeration is complete.

D. Finite word families

The remaining truncation is over source words, and it is organized so that contributions provably never overlap. For a base set S of run lengths and word length $L = m + 2R$, the *full family* is S^L ; for $x \notin S$, the *one-extra family* $\mathcal{F}(S; x)$ consists of words with exactly one coordinate equal to x and the rest in S . Within a fixed (m, R) : distinct one-extra lengths give disjoint families, all disjoint from the full family. Across different R the word lengths differ. Across different m the certified quantities are different increments Δ_m , so no disjointness is needed at all: the hierarchy (57) adds them as separate nonnegative terms. Appendix C records these observations formally. The value of this organization is that every enlargement of a family—more base lengths, more extra lengths, larger R , longer windows—adds provably nonnegative certified mass, so partial computations compose.

VIII. DIRECTED FIXED-POINT CERTIFICATION

The bounds of Sections VI and VII leave finite sums of terms of the shape

$$p(w) \times (\text{combination of } F \text{ values and integrals}) \quad (94)$$

at integer weights. This section specifies the arithmetic that evaluates such terms and proves that every rounding acts toward the final inequality. The guiding rule: *multiplicities stay exact integers; probabilities are rounded down; logarithms are rounded in the direction their sign requires.*

A. Source-word probabilities

Lemma 23 (Probability floor). *Let the witness probabilities be dyadic, $P_x = w_x/2^{30}$. Then $p(w)$ of (62) is exactly dyadic, and replacing it by*

$$\underline{p}(w) = 2^{-224} \lfloor 2^{224} p(w) \rfloor \leq p(w) \quad (95)$$

in any certified sum can only decrease the sum. Words whose $\underline{p}(w)$ underflows to zero may be discarded.

Proof. Each retained pointwise quantity multiplying $p(w)$ —a fine-path F term, a clipped correction bracket, a clipped 2×2 mutual-information block, a midpoint sum—is nonnegative by construction. A sum $\sum_w p(w) c_w$ with $c_w \geq 0$ is nondecreasing in each $p(w)$. $\square$

B. Directed binary logarithms

Lemma 24 (Fixed-point $\log_2$ bounds). *For every rational $r = n_1/n_0 \geq 1$ represented by positive integers $n_0, n_1 < 2^{128}$ (the scaled arithmetic range used below), there are computable $\underline{\log}(r), \overline{\log}(r) \in 2^{-112}\mathbb{Z}$ with*

$$\underline{\log}(r) \leq \log_2 r \leq \overline{\log}(r), \quad (96)$$

obtained as follows. Choose the integer e with $z = r2^{-e} \in [1, 2)$ and set $y = (z - 1)/(z + 1) \in [0, 1/3]$. Then

$$\log_2 r = e + \frac{2}{\ln 2} \sum_{n=8}^{\infty} \frac{y^{2n+1}}{2n+1}, \quad (97)$$

a series with nonnegative terms. The lower routine truncates after eight terms, rounds y and every intermediate product down, and multiplies by

$$K_{lo} = 53226052391377289966 \leq 2^{64} \frac{2}{\ln 2}; \quad (98)$$

the upper routine rounds upward, adds the tail bound

$$\sum_{n=8}^{\infty} \frac{y^{2n+1}}{2n+1} \leq \frac{y^{17}}{17(1-y^2)} \leq \frac{9y^{17}}{136}, \quad (99)$$

and multiplies by

$$K_{up} = 53226052391377289967 > 2^{64} \frac{2}{\ln 2}. \quad (100)$$

Proof. Equation (97) is the identity $\ln z = 2 \operatorname{artanh} y$ with $y = (z - 1)/(z + 1)$, divided by $\ln 2$; all series terms are nonnegative on $y \in [0, 1/3]$, so truncation lowers the value and downward rounding of y lowers it further, each partial sum being a polynomial with nonnegative coefficients. For the tail, $\sum_{n \geq 8} \frac{y^{2n+1}}{2n+1} \leq \frac{1}{17} \sum_{n \geq 8} y^{2n+1} = \frac{y^{17}}{17(1-y^2)}$, and $1 - y^2 \geq 8/9$ on $[0, 1/3]$ gives the final form of (99). Exact rational bounds from the same atanh series give $K_{lo} \ln 2 \leq 2^{65} < K_{up} \ln 2$, $\ln 2 < 355/512$, and $(1 - y^2)^{-1} \leq 9/8$. Multiplication by K_{lo} (resp. K_{up}), followed by a downward (resp. upward) integer shift, therefore preserves the required direction. All returned values are multiples of 2^{-112} . $\square$

C. Directed evaluation of F

Lemma 25 (Scaled F bounds). *Let a, b be nonnegative integers, possibly wider than the logarithm input, and let $s \geq 0$ be a common shift. Then*

$$2^s F(\lfloor a/2^s \rfloor, \lfloor b/2^s \rfloor) \leq F(a, b), \quad (101)$$

$$F(a, b) \leq 2^s F(\lceil a/2^s \rceil, \lceil b/2^s \rceil), \quad (102)$$

Write $\log_- := \underline{\log}$, $\log_+ := \overline{\log}$ and, for $x, y > 0$,

$$L_\sigma(x, y) = x \log_\sigma \frac{x+y}{x} + y \log_\sigma \frac{x+y}{y}, \text{quad } \sigma \in \{-, +\}. \quad (103)$$

Choose s so that each sum of scaled arguments is below 2^{128} . Then

$$\underline{F} = 2^s L_-(\lfloor a/2^s \rfloor, \lfloor b/2^s \rfloor), \quad \overline{F} = 2^s L_+(\lceil a/2^s \rceil, \lceil b/2^s \rceil) \quad (104)$$

satisfy $\underline{F} \leq F(a, b) \leq \overline{F}$. Set $\underline{F} = 0$ if a floored argument vanishes, and set all three quantities to zero if an original argument vanishes.

Proof. Homogeneity gives $F(a, b) = 2^s F(a/2^s, b/2^s)$, and F is nondecreasing in each argument (Lemma 15), so replacing the scaled arguments by their floors (ceilings) moves the value down (up). For positive arguments use the cancellation-free identity

$$F(a', b') = a' \log_2 \frac{a' + b'}{a'} + b' \log_2 \frac{a' + b'}{b'}. \quad (105)$$

Both coefficients and both logarithms are nonnegative, so the directed logarithms give (104); the zero convention follows from $F(a, 0) = F(0, b) = 0$. The profile sums use the equivalent chord form of Appendix A, whose ordinates are certified below the concave normalized function. $\square$

D. Leaf integrals by midpoint rule

Lemma 26 (Convex midpoint lower bound). *For $c > 0$ the integrand $f(x) = \log_2(1 + c/x)$ is convex on $(0, \infty)$:*

$$f''(x) = \frac{c(2x+c)}{\ln 2 x^2(x+c)^2} > 0. \quad (106)$$

Hence for $b \geq 0$, $a > 0$, and any $q \in \mathbb{N}$,

$$\int_b^{b+a} f(x) dx \geq \frac{a}{q} \sum_{j=0}^{q-1} f\left(b + \frac{j+\frac{1}{2}}{q} a\right), \quad (107)$$

Let $b' = \lceil b/2^s \rceil$, $a' = \lfloor a/2^s \rfloor$, and $c' = \lfloor c/2^s \rfloor$. Applying the right side to (b', a', c') , evaluating each logarithm with $\log$, and multiplying the result by 2^s remains a lower bound for the original integral.

Proof. Differentiate twice; positivity of f'' is clear from the displayed formula. On each subinterval I_j of length a/q , Jensen's inequality for the uniform measure on I_j gives $f(\text{midpoint of } I_j) = f(\mathbb{E}U) \leq \mathbb{E}f(U) = \frac{a}{q} \int_{I_j} f$, and summing proves (107); at $b = 0$ the integral is finite (Lemma 18) and all midpoint arguments are positive.

For the rounding step, write $I(b, a, c) = \int_b^{b+a} \log_2(1 + c/x) dx$. The integral is nonincreasing in b , nondecreasing in a and c , and homogeneous: $I(\gamma b, \gamma a, \gamma c) = \gamma I(b, a, c)$. Therefore

$$I(b, a, c) \geq 2^s I(b', a', c'). \quad (108)$$

Applying (107) to the rounded integral and then replacing every logarithm by $\log$ proves the claimed bound. This comparison is made at the integral level and does not assume a separate termwise monotonicity of the midpoint sum in a . $\square$

E. Arithmetic scales

The finite-window evaluations round probabilities on the 2^{-224} grid (Lemma 23) and entropic factors on the 2^{-112} grid (Lemma 24); each retained subtotal is therefore a nonnegative integer divided by

$$2^{224+112} = 2^{336}. \quad (109)$$

The adjacent-path profile evaluation uses 2^{-160} for each of two profile probabilities and 2^{-96} for chord ordinates, hence denominator $2^{2 \cdot 160 + 96} = 2^{416}$ (Appendix A). The moments μ, D, λ , every subtotal, and the final comparison are then combined as exact rational numbers. Floating-point values are used only for the descriptive plots and independent numerical cross-checks, never for the inequalities in Theorem 27.

IX. NUMERICAL VERIFICATION

All analytical identities and entropy inequalities up to this point hold for an arbitrary finite-support run-length law; Lemma 23 has already stated the additional dyadic assumption used by the arithmetic instantiation. What remains is to choose a witness P , select finite instantiations of the bounds—a path depth $R_{\max}$, source-word families, and midpoint counts—and evaluate the resulting finite sums with the directed arithmetic of Section VIII. The numerical work does not supply an additional inequality: it evaluates expressions already proved to be lower bounds. We report both the directed rational values used in the theorem and independent high-precision checks on small instances.

TABLE I
LARGEST PROBABILITIES IN THE DYADIC RUN-LENGTH WITNESS

x	P_x	x	P_x
3	0.315515902	2	0.201070396
7	0.144942431	13	0.108788959
8	0.078570154	20	0.068446696
28	0.031053787	14	0.017304989
37	0.009520105	36	0.005427203

A. Dyadic witness

The witness has 84 positive masses on $\{1, \dots, 150\}$ and probabilities

$$P_x = \frac{w_x}{2^{30}}, \quad \sum_{x=1}^{150} w_x = 2^{30}, \quad (110)$$

The complete list of nonzero integers w_x is given in Appendix D; all omitted weights are zero. The law was found by numerical optimization of the single-run objective followed by dyadic quantization. Its origin is irrelevant to the proof, which treats the displayed integers as constants. Its exact moments are

$$\mu = 8.325190612114965915679931640625, \quad (111)$$

$$D = 0.091160767387253067893770010323137978055 \dots, \quad (112)$$

$$\lambda = 0.10004701554492413709314611843033461210 \dots \quad (113)$$

Table I shows the largest masses for orientation; every computation uses the full witness.

B. Single-run baseline

The baseline is evaluated directly from Fact 9: truncate the nonnegative series (39) at $i = 6$, compute the first 200 coefficients of (38) exactly, merge the remaining probability into one atom, and use the lower logarithm bounds of Lemma 24. This reproduces (41) and hence proves $R_{1\text{run}}(P) > 0.10280005$.

C. Three-run instantiation

1) *Adjacent path:* Proposition 19 is instantiated with $R_{\max} = 5$. The profile method of Appendix A evaluates finite sub-support lower bounds $\mathcal{J}_{s,i}$ for all fifteen pairs (s, i) , restricting the active profile coordinates to the 30, 20, 12, 10, 6 most probable run lengths for $s = 1, \dots, 5$, respectively; shared fully deleted coordinates are summed exactly through the factor D^{s-1} . By Proposition 19 each restriction is safe. The remaining profile sum is evaluated on the 2^{-416} grid. Substituting the directed values into (81) gives

$$g_{3,\text{path}} > 0.05580644032938115044064563577540621 \dots, \quad (114)$$

with the five R -level subtotals of Table II.

TABLE II
CERTIFIED THREE-RUN ADJACENT-PATH CONTRIBUTIONS

R	Contribution (bits/output run)
1	0.040871046886
2	0.012215987150
3	0.002318104252
4	0.000357684791
5	0.000043617248
Total	0.055806440329

2) *Endpoint corrections*: The corrections (84) are evaluated on the following disjoint word families (Section VII-D), with midpoint counts $q = 8$ for $R = 1$ and $q = 4$ otherwise: $R = 1$: the full family over the nine most probable lengths $\{3, 2, 7, 13, 8, 20, 28, 14, 37\}$ and the one-extra families for lengths 36, 27, 46, 45; $R = 2$: the full family over $\{3, 2, 7, 13\}$ and the one-extra families for 8, 20, 28, 14, 37, 36; $R = 3$: the full family over $\{3, 2, 7\}$. For $R = 1$ the evaluation retains a common-support rectangular subrange of output lengths and discards the remaining nonnegative terms. For $R = 2, 3$ it uses the exhaustive second-largest recursion of Lemma 22. Thus no exhaustiveness claim is needed for the $R = 1$ contribution. The certified totals are

$$\begin{aligned} \mathbb{E}[\Gamma_1] &> 0.01126783205429382752389036830045582\dots, \\ \mathbb{E}[\Gamma_2] &> 0.002441921733798230318109114198972588\dots, \\ \mathbb{E}[\Gamma_3] &> 0.00007726347089112966913350995389374416\dots \end{aligned} \quad (115)$$

Adding (114) and (115),

$$g_3 > 0.0695934575883643379517786282287283596. \quad (116)$$

D. Higher-window instantiation

The increments $\Delta_4, \dots, \Delta_8$ are certified through Lemmas 20–22 on the families of Table III. Within each (m, R) the families are pairwise disjoint; across m they feed different increments and add by (57).

Summing the exact directed rational values within each window,

$$\begin{aligned} \Delta_4 &> 0.004296697091384994126972525365333410\dots, \\ \Delta_5 &> 0.0009959163756625601805375614645531863\dots, \\ \Delta_6 &> 0.0002578260279368504506026011281440604\dots, \\ \Delta_7 &> 0.00005679701748214544361806241752875797\dots, \\ \Delta_8 &> 0.000004948753610993626861817622252613214\dots \end{aligned} \quad (117)$$

With (116) and (57),

$$e(P) \geq g_8 > 0.0752056428544418817804. \quad (118)$$

No claim is made that these families exhaust any Δ_m ; every excluded observation is counted as zero.

E. Independent small-instance cross-checks

As an independent numerical check of the directed formulas, the all-2 source word in the witness was evaluated by a direct 100-digit enumeration of every segmentation and output-length vector. Table IV compares those reference values with the directed lower evaluations. The direct values are not used in the proof; their role is to test the combinatorial enumeration and the direction of rounding. In all six cases the directed result lies below the independently evaluated value, with the larger $R = 2$ three-run gap explained by the deliberately coarse four-panel midpoint rule.

F. Main result

Table V collects every additive term. Let $\underline{g}_8$ denote the exact rational sum of the directed values represented by its final row. The displayed decimals are shortened downward; the theorem uses $\underline{g}_8$ itself, not those decimals.

Define the exact rational certificate

$$\begin{aligned} \underline{\mathcal{R}} &:= \frac{10280005}{10^8} + \lambda \underline{g}_8 \\ &= 0.11032415011972435980010110\dots \end{aligned} \quad (119)$$

The directed bounds give $\mathcal{R}(P) > \underline{\mathcal{R}}$.

Theorem 27 (Computer-assisted half-deletion lower bound). *The capacity of the binary deletion channel at deletion probability one half satisfies*

$$C(1/2) > 0.11032415. \quad (120)$$

Proof. By Theorem 8, $C(1/2) \geq R_{\text{run}}(P) + \lambda e(P)$ for the witness P . The baseline inequality $R_{\text{run}}(P) > 0.10280005$ follows from the direct finite evaluation of Fact 9 with the safe truncations justified in Section III-C. Theorem 11 gives $e(P) \geq g_8$; Proposition 19, the exact split (87), and the increments certified through Lemmas 20–22 give $g_8 \geq \underline{g}_8$, with every contribution rounded toward zero from below by the directed arithmetic proved correct in Section VIII. Thus $C(1/2) > \underline{\mathcal{R}}$. The final exact-rational comparison is

$$\underline{\mathcal{R}} - \frac{11032415}{10^8} > 0, \quad (121)$$

which is evaluated between the rational numbers constructed by the directed finite procedure; the displayed decimals are not used in its sign test. $\square$

Figure 4 shows how the certified rate accumulates as successive nonnegative terms are included. It also separates the numerical optimization that produced the single-run baseline from the theoretically justified endpoint-window corrections. The plotted decimals are descriptive and take no part in the proof.

G. Guarded public upper-bound reconstruction

For context we restate the strongest public upper bound as an exact rational statement. Let $C_{n,k}$ denote the capacity of the

TABLE III
FINITE SOURCE-WORD FAMILIES USED FOR THE HIGHER-WINDOW INCREMENTS

Window m	R	Source-word family	Certified contribution (bits/output run)
4	1	full {3, 2, 7, 13, 8, 20}	0.003142892114
4	1	one extra 14, 28, or 37; remainder in the preceding set	0.000626147753
4	2	full {3, 2, 7}	0.000223380610
4	2	one extra 8 or 13; remainder in {3, 2, 7}	0.000278424540
4	3	full {3, 2}	0.000006913825
4	3	one extra 7 or 13; remainder in {3, 2}	0.000018938247
5	1	full {3, 2, 7, 13, 8}	0.000972048190
5	2	full {3, 2}	0.000023868184
6	1	full {3, 2, 7, 8}	0.000248702939
6	2	full {3, 2}	0.000009123087
7	1	full {3, 2, 7}	0.000056797017
8	1	full {3, 2}	0.000004948753

TABLE IV
INDEPENDENT DIRECT-ENUMERATION CROSS-CHECKS ON THE ALL-2 SOURCE WORD

Quantity	R	Directed lower value	Direct value	Difference
Three-run endpoint correction	1	$1.467370192240681 \times 10^{-5}$	$1.467370192395545 \times 10^{-5}$	1.55×10^{-15}
Three-run endpoint correction	2	$4.861917810655579 \times 10^{-7}$	$4.862017982745881 \times 10^{-7}$	1.00×10^{-11}
Four-run endpoint CMI	1	$1.476807194956723 \times 10^{-6}$	$1.476807195003277 \times 10^{-6}$	4.66×10^{-17}
Four-run endpoint CMI	2	$5.453002623730097 \times 10^{-8}$	$5.453002626862988 \times 10^{-8}$	3.13×10^{-17}
Five-run endpoint CMI	1	$1.669573508798467 \times 10^{-7}$	$1.669573508898811 \times 10^{-7}$	1.00×10^{-17}
Five-run endpoint CMI	2	$7.153477543464354 \times 10^{-9}$	$7.153477548629310 \times 10^{-9}$	5.16×10^{-18}

TABLE V
CERTIFIED BOUNDARY-ENTROPY LEDGER

Component	Bits/output run
Three-run fine path	0.055806440329
$R = 1$ endpoint correction	0.011267832054
$R = 2$ endpoint correction	0.002441921733
$R = 3$ endpoint correction	0.000077263470
g_3 lower bound	0.069593457588
Δ_4	0.004296697091
Δ_5	0.000995916375
Δ_6	0.000257826027
Δ_7	0.000056797017
Δ_8	0.000004948753
g_8 lower bound	0.075205642854

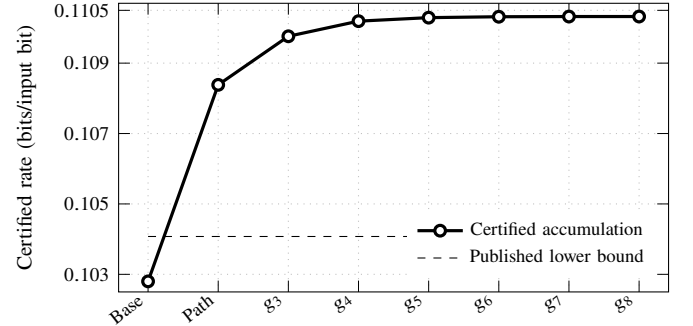


Fig. 4. Cumulative certified lower bound on $C(1/2)$. After the numerically optimized single-run baseline, every step adds a theoretically nonnegative term: the three-run path, the endpoint corrections completing g_3 , and $\Delta_4, \dots, \Delta_8$. The vertical scale is zoomed; the theorem uses the exact rational ledger.

channel mapping n input bits to a uniformly random length- k subsequence. The finite-channel inequality of [5] (building on [10]) gives

$$C(d) \leq \frac{1}{n} \sum_{k=0}^n \binom{n}{k} d^{n-k} (1-d)^k C_{n,k}. \quad (122)$$

At $d = 1/2$, $n = 29$, substituting the four-decimal upper values of $C_{29,k}$ printed in [5] yields the exact rational

$$U_{\text{print}} = 0.189535319340042769908905029296875 \dots \quad (123)$$

To be independent of whether each printed decimal was rounded or truncated, a full unit of the last printed decimal, 10^{-4} , is added to every $k \geq 1$ entry; the guard totals $\delta_U = \frac{10^{-4}}{29}(1 - 2^{-29})$, and

$$U_{\text{print}} + \delta_U = 0.18953876761589841596011458 \dots < 0.189538767616. \quad (124)$$

This proves (7) under the sole assumption that each printed table entry understates its true value by less than one unit in the last printed decimal. It is a conservative restatement of published data, not a new computation; together with Theorem 27 it yields the interval (8), of width less than 0.0792147.

X. DISCUSSION

Scope. We import only the renewal information-rate theorem (Fact 7) and the single-run formula (Fact 9); the boundary-variable correspondence is proved in (36), and all other analytical steps are proved here. Theorem 27 is computer assisted: deterministic finite sums and exact rational comparisons replace statistical evidence. Their implementation remains part of the review surface; Table IV is an independent check, not a formal verification.

Lower-side headroom. The certificate truncates the path at $R = 5$, the four-run term at $R = 3$, the five- and six-run terms at $R = 2$, the seven- and eight-run terms at $R = 1$, and the hierarchy at $M = 8$. Every discarded term is nonnegative, so larger families or windows improve the bound monotonically. The decay in Table V suggests saturation near the 10^{-4} scale for this witness, but only the displayed lower bound is certified. Moreover, the posterior weights are independent of P and d ; at $d = 1/2$ each finite family depends on P through monomials $\prod_x P_x 2^{-x}$. Thus dyadic laws could be optimized by interval branch-and-bound without recomputing the combinatorial coefficients.

Upper side. Improvement requires a new finite-channel certificate, not another mixture of the separately optimized printed $C_{n,k}$. One candidate is the common-input information-radius dual

$$\min_{Q_0, \dots, Q_n} \max_{x \in \{0,1\}^n} \sum_{k=0}^n 2^{-n} \binom{n}{k} D(W_{n,k}(\cdot | x) \| Q_k), \quad (125)$$

which cannot exceed the per-output-length mixture in (122). Certifying it would require a separation oracle for the maximizing word and interval bounds on the divergences, independently of the present lower-bound theory.

Beyond i.i.d. runs. Dependent run processes may improve the rate, but require a new regenerative hierarchy. The reusable template is to isolate a nonnegative correction, approximate it by monotone finite windows, integerize the posterior, and certify finite pieces by directed arithmetic.

XI. CONCLUSION

The hidden segmentation term in the exact renewal information rate of the binary deletion channel is the increasing limit of finite endpoint-window entropies, and each window enlargement contributes an explicit nonnegative conditional mutual information. The segmentation posterior is integer weighted for every deletion probability; at one half a dyadic run-length law makes its common factor dyadic. Entropy-tree bounds, leaf-absorption integrals, log-sum corrections, binary chain rules, and directed fixed-point arithmetic then convert finitely many windows into a deterministic certificate. For the explicit witness of Appendix D, the certificate proves $C(1/2) > 0.11032415$. A guarded reconstruction of the strongest published finite-channel table places the capacity below 0.189538767616 under the table-precision assumption stated in Section IX-G.

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APPENDIX A

THE PROFILE ALGORITHM FOR $\mathcal{J}_{s,i}$

This appendix connects the direct sum (80) to the profile algorithm used in its finite evaluation. Fix (s, i) and split the local word at the two cuts into a left part (the coordinates entering $t_L^{(i)}$ or $t_L^{(i+1)}$) and a right part. For each output-length pair, the two multiplicity products in (80) factor into one left and one right profile entry, so each term has the form

$$F(X_L X_R, Y_L Y_R) \quad (126)$$

with nonnegative integers and with dyadic probability weights attached separately to the two sides. By homogeneity, with $\rho = Y_L Y_R / (X_L X_R) \leq 1$ (the other case by symmetry),

$$F(X_L X_R, Y_L Y_R) = X_L X_R g(\rho), \quad g(\rho) = (1+\rho) \log_2(1+\rho) - \rho \log_2 \rho. \quad (127)$$

The evaluation partitions $[0, 1]$ into 64 rational intervals and replaces the concave function g by dyadic chords lying below it; on each interval the bound $g(\rho) \geq \alpha + \beta \rho$ with certified $\alpha, \beta \geq 0$ turns the double profile sum into monotone product-threshold queries. Sorting both profiles by Y/X and sweeping cumulative sums evaluates the complete finite sub-support sum without enumerating all profile pairs. Left and right profile probabilities are rounded down to 2^{-160} and chord ordinates to 2^{-96} , giving the 2^{-416} scale stated in Section VIII. Every chord ordinate is reconstructed from rational atanh bounds (Lemma 24), and chordwise validity and nonnegative coefficients are checked exactly. The selected source-length counts per s are 30, 20, 12, 10, 6; by Proposition 19 the restriction discards only nonnegative terms.

APPENDIX B

ENDPOINT-CORRECTION BOOKKEEPING

This appendix records how (84) is evaluated pointwise, in the notation of Section VI-C. Let $z_{a,c}$ be as in (76) and $u_c = \sum_a z_{a,c}$; set

$$P_{\text{fine}} = \sum_{c=1}^R \sum_a F(z_{a,c-1}, z_{a,c}), \quad P_{\text{coarse}} = \sum_{c=1}^R F(u_{c-1}, u_c). \quad (128)$$

Adding and subtracting P_{coarse} ,

$$\mathcal{H}(u) - P_{\text{fine}} = \underbrace{[\mathcal{H}(u) - P_{\text{coarse}}]}_{\Gamma_R^{\text{tree}}} + \underbrace{[P_{\text{coarse}} - P_{\text{fine}}]}_{\Gamma_R^{\text{logsum}}}, \quad (129)$$

which is (87); nonnegativity of the two brackets follows from Lemmas 17 and 16, valid verbatim when some weights vanish since $F(0, \cdot) = 0$.

For Γ_R^{tree} , absorb the path’s leaves one at a time. Lemma 18 shows each absorption removes exactly the incident edge term (identity (73)) and credits the change of the surviving incident edge as the integral (75); iterating to a single vertex accounts for the whole gap, not a subset. On a path each absorption has at most one surviving incident edge, so the gap becomes a sum of at most $R - 1$ integrals per absorption order. Four deterministic absorption orders are evaluated and the pointwise maximum of the four certified lower bounds is retained; a

TABLE VI
NONZERO NUMERATORS w_x OF THE DYADIC WITNESS $P_x = w_x/2^{30}$

x	w_x	x	w_x	x	w_x	x	w_x
2	215,897,694	28	33,343,750	49	103,015	70	5,665
3	338,782,620	29	195,690	50	76,560	71	2,970
7	155,630,750	30	67,925	51	82,995	72	1,870
8	84,364,060	31	43,725	52	115,610	73	1,320
11	990	32	41,360	53	200,475	74	1,045
12	3,080	33	56,045	54	484,000	75	935
13	116,811,255	34	118,415	55	1,360,370	76	935
14	18,581,090	35	517,660	56	1,266,760	77	1,045
15	8,525	36	5,827,415	57	221,815	78	1,705
16	5,170	37	10,222,135	58	54,945	79	2,640
17	5,940	38	682,385	59	25,025	80	5,335
18	11,550	39	207,955	60	15,950	81	122,375
19	39,655	40	118,745	61	12,760	82	507,210
20	73,494,080	41	99,605	62	12,430	83	2,530
21	54,395	42	119,295	63	15,070	100	171,160
22	27,115	43	208,230	64	24,145	120	1,595
23	17,270	44	553,135	65	41,635	121	5,500
24	17,985	45	1,880,065	66	168,245	122	18,040
25	30,250	46	3,430,900	67	1,380,060	123	4,455
26	115,390	47	1,183,435	68	129,800	124	1,430
27	4,012,525	48	241,450	69	15,565	150	4,125

maximum of lower bounds on the same quantity is a lower bound. Each integral is certified by Lemma 26.

For Γ_R^{logsum} the directed arithmetic evaluates $\underline{F}(u_{c-1}, u_c) - \sum_a \bar{F}(z_{a,c-1}, z_{a,c})$ per bracket (Lemma 25) and clips negative results to zero. Summing the two certified parts over a finite word family, with probabilities floored per Lemma 23, gives (115). The $R = 2, 3$ evaluations use the exhaustive recursion of Lemma 22; the $R = 1$ evaluation retains only the common-support output rectangle and safely discards the rest.

APPENDIX C EVENT-FAMILY DISJOINTNESS

A source word of an m -window with total parameter R has $m + 2R$ coordinates. For a base set S , the full family is S^{m+2R} ; for $x \notin S$,

$$\mathcal{F}(S; x) = \{w : \text{exactly one coordinate equals } x, \\ \text{all other coordinates lie in } S\}. \quad (130)$$

If $x \neq y$ are both outside S , then $\mathcal{F}(S; x) \cap \mathcal{F}(S; y) = \emptyset$ and both are disjoint from S^{m+2R} : membership prescribes the multiset of coordinates outside S . The instantiation uses only unions of this form within a fixed (m, R) . Families at different R concern different word lengths, hence disjoint observation events within the same window quantity. Families at different m enter different nonnegative increments Δ_m of the exact telescoping (57), so no cross- m disjointness is required: reusing the same run lengths at several m is sound because the certified quantities, not the events, are distinct.

APPENDIX D COMPLETE DYADIC WITNESS

Table VI lists every nonzero numerator in (110); all unlisted w_x for $1 \leq x \leq 150$ are zero. The 84 displayed integers sum to $2^{30} = 1073741824$, so they define the probability law used throughout Section IX without reference to external data.

APPENDIX E EXACT NUMERICAL LEDGER

The underlined symbols below denote the exact directed rational values, whose decimal expansions are

$$g_{3,\text{path}} = 0.05580644032938115044 \dots,$$

$$\mathbb{E}[\Gamma_1] = 0.01126783205429382752 \dots,$$

$$\mathbb{E}[\Gamma_2] = 0.00244192173379823032 \dots,$$

$$\mathbb{E}[\Gamma_3] = 0.00007726347089112967 \dots,$$

$$\underline{g}_3 = 0.06959345758836433795 \dots,$$

$$\underline{g}_8 = 0.07520564285444188178 \dots,$$

$$\lambda \underline{g}_8 = 0.00752410011972435980 \dots,$$

$$\underline{\mathcal{R}} := \frac{10280005}{10^8} + \lambda \underline{g}_8,$$

$$\underline{\mathcal{R}} = 0.11032415011972435980010110 \dots$$

The final comparisons are the exact rational assertions

$$\underline{\mathcal{R}} - \frac{11032415}{10^8} > 0, \quad (131)$$

$$\frac{1}{29 \cdot 2^{29}} \sum_{k=1}^{29} \binom{29}{k} \hat{C}_{29,k} + \frac{1 - 2^{-29}}{29 \cdot 10^4} < \frac{189538767616}{10^{12}}, \quad (132)$$

where $\hat{C}_{29,k}$ are the printed table values of [5] as exact decimal fractions.

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